Impact of Macroeconomic Landscape on Academic Pursuit

Aroonrassamee Wongkeaitaroon

Economics Department, San Jose State University

ECON103A: Introduction to Econometrics

Dr. Sanchita Mukherjee

November 15, 2025

Abstract

The fluctuations of post-secondary education enrollments have been a topic of discussion in the past half-century. Past research has documented the countercyclical trends of undergraduate program enrollment and economic downturn, but has only specified bivariate models with unemployment as the only independent variable. We attempt to model the larger macroeconomic landscapes and their effects on both undergraduate and graduate program enrollments from 1986 to 2023. Thereby, we conclude that graduate enrollments are less correlated with macroeconomic measures than undergraduate enrollments; ergo, graduate program enrollments are less sensitive to macroeconomic changes.

1. Introduction

Over the past couple of decades, the rise of post-secondary education has gained popularity. Between 1993 and 2023, 74.9% increase in college graduates (Hanson, 2025). Although overall college attainment has been on a steady rise, data from the US Bureau of Labor Statistics show a fluctuating pattern of college enrollment rates of high school graduates from 1993 to now (Bureau of Labor Statistics, 2024). The oscillating rates indicate that the motivation for high school graduation to enroll in college immediately may be indicative of other factors beyond simple educational demand. Simultaneously, the percentage of jobs that require their workers to have at least some college, bachelor's degrees, and master's degrees is on the rise (Georgetown University Center on Education and the Workforce, 2024). Yet, enrollment numbers are not reflective of the changes in these percentages, highlighting a potential divergence between workforce demands and educational participation.

The unpredictable nature of college enrollment gives rise to an important question: What macroeconomic changes motivate or discourage individuals in the U.S. population from pursuing

post-secondary education? This paper aims to examine the macroeconomic influence on students' motivation to enroll in higher education by controlling for parameters such as GDP per capita, unemployment, real interest rates, and the cost of attending college. Furthermore, it investigates the same parameters on both graduate school enrollment and undergraduate enrollment to see how sensitive graduate school enrollment is to undergraduate school enrollment and to macroeconomic variables.

2. Literature Review

The trend between them is an interesting topic and has been extensively discussed during the past couple of decades. The Education Advisory Board's article *How to Understand Higher Education Enrollments in Economic Recessions* observed the countercyclical trends between the economy and higher education enrollments. The infographic analyzed the trends of vocational, community college, undergraduate, and graduate programs enrollment superposed on the unemployment trend between 2006 and 2016. From their observation, they claimed that "recession increased graduate enrollments only slightly… the impact was small because most unemployed workers in the last recession lacked the bachelor's degrees necessary to enroll in a graduate program. Additionally, they observed that the growth in traditional students (under the age of 25) did not change significantly during the 2007-2011 recession (Education Advisory Board, 2016). While these are interesting and important observations, the bivariate model did not suffice for the enrollment trend. This limitation can be fixed by a multivariate model, as presented in this paper, to better identify the drivers of enrollment behavior

Simultaneously, Georgetown University Center on Education and the Workforce observed that the percentage of jobs that will require their workers to hold a degree is on the rise. In 1983, 11% of jobs required a bachelor's degree; in 2021, this number is 22%, and in 2031, the

projected percentage is 16%. In parallel, the percentage of jobs that require at least a graduate degree was 8% in 1983, 14% in 2021, and projected to be 16% in 2031. These are the total average percentages, which do not account for industry differences. Of course, the professional and managerial sector will require higher education for the average worker than for blue-collar professions.

A report by the TIAA Institute*, Macro-Trends Impacting the Evolution of Higher Education: How Colleges and Universities Can Adapt to Create Value in the 21st Century*, provides key insights into how institutions have been impacted by the economic transformation towards increasingly skill-based labor. Through empirical analyses, they have found that the cost of college increased 143% between 2000 and 2022, significantly outpacing tuition discounting, and financial aid also rose steadily over that period. At the same time, state governments' financial support for higher education has declined since 1990, falling from 12% of the state budgets to under 9%. Beyond these trends, the report cited that while skepticism of the continuing value of college education has been increasing, a college degree provides benefits to both the community, such as lowering incarceration rates and dependence on public assistance, and the individual pursuing a college education, in better health, improved longevity, and greater happiness. The report further proposes strategies that institutions can implement to encourage more people to pursue higher education: support students through graduation, demystify college costs, embrace online education, adapt to the changing labor market, and provide lifelong learning. While this report is comprehensive and insightful, the primary focus lies on institutional and student-level perspectives rather than the macroeconomy, and it does not provide statistical analyses to support the correlation between the macro-trends and enrollments.

Many reports also observed trends in enrollment across demographic groups, such as gender, race and ethnicity, family income, and age. For example, Deloitte Insights' *Trend No. 1: College Enrollment Reaches Its Peak* (2023) cited that the recent decade's decline in college enrollment has been most prevalent among men. "Women surpassed men in terms of mean years of schooling in the 1980s, and now female students account for 56% of undergraduates enrolled at US institutions". This trend has rippled its effect into the labor market, with the employment of working-aged men reaching Depression-era levels. On the other hand, students of color and low-income students are much more likely to either not start college or stop without a degree. The article calls for the institution to improve access to higher education by updating academic programs and recruitment strategies. Although this report offers insights that align with the broader conversations about changes in enrollment trends, its focus is not within the scope of macroeconomic analysis presented in this paper.

Together, these studies highlight several important themes relating to post-secondary higher education enrollment: a growing divergence between labor-market demand for higher education and actual enrollment trends, the difference in enrollment by demographics, and offer institutions with potential policy changes to help keep up with the economy. However, while reviewing the relevant literature provides valuable insights, many do not present sufficient rigorous statistical analyses, which directly link macroeconomic conditions with enrollment trends. The multivariate approach used in this paper helps fill the gap by demonstrating empirical factors such as unemployment, real interest rates, change in GDP per capita, and the average cost of obtaining graduate and undergraduate degrees

3. Model Specification

3.1 Economic Model

To compare the effects and sensitivity of macroeconomic parameters on undergraduate vs graduate programs, we define a generalizable model. We adopt the following model for both graduate and undergraduate enrollments:

Equation 1: Student Enrollment Multivariate Ordinary Least Squares Model

$$Enrollment_t = \beta_0 + \beta_1 GDP_t^2 + \beta_2 GDP_t + \beta_3 Cost + \beta_4 Unemployment_t + \beta_5 RIR + \epsilon_t$$

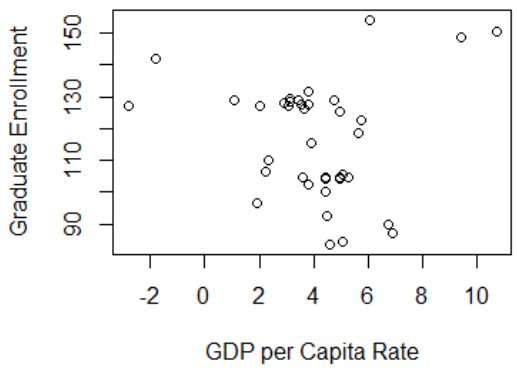
3.2 Variable Explanation

Table 1: Codebook for Enrollment and Macroeconomic Parameters

Variable Name	Description	Type	Year	Source
gradenroll	Graduate Enrollment	Dependent	1986-2023	NCES
ugenroll	Undergraduate Enrollment	Dependent	1986-2023	NCES
capitarate	GDP per capita rate of change	Independent	1986-2023	World Bank
gradcost	Average tuition and required fees of a graduate program per year.	Independent	1989-2023	NCES
undergradcost	Average tuition and required fees of an undergraduate program per year.	Independent	1986-2022	NCES
unemployment	Unemployment	Independent	1986-2023	FRED
RIR	Real Interest Rate	Independent	1986-2023	FRED

The enrollments are undergraduate and graduate enrollments each year from 1986 to 2023, measured in thousands (NCES, 2024). The dependent variable is enrollment rather than participation, attendance, or admitted students because we are measuring the macroeconomic

conditions that motivate students to attempt to attend post-secondary education, whether they are accepted by the institution.



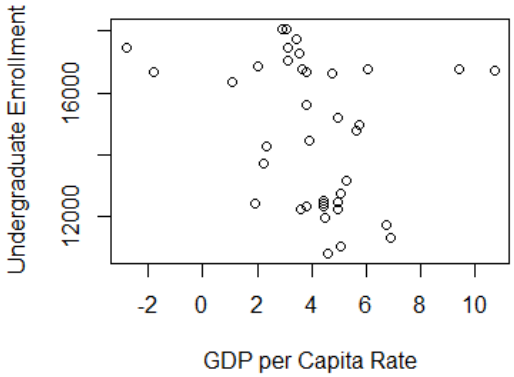


Figure 1 (left) Quadratic relationship between GDP per Capita Rate and Graduate Enrollment. Figure 2 (right) Quadratic relationship between GDP per Capita Rate and Undergraduate Enrollment.

The GDP per capita rate of change, GDP_t (World Bank, 2024), is part of the model to examine how changes in a country's economic output impact the economic climate and motivate the collective population to pursue higher education. GDP per capita is chosen over GDP to better observe the macroeconomic impact on a smaller scale, better representing the change in the standard of living and economic well-being of the citizens. As seen in Figure 1, the relationship between GDP per capita and graduate enrollment is quadratic. Therefore, in the model, both the GDP per capita rate of change (capitarate) and GDP rate of change squared (capitasquare) are included to increase the explanatory power of the variable. Furthermore, squaring the rate of change captures the nonlinear expectation that large economic change, either rapid growth or decline, can heighten opportunity costs or create financial uncertainty, which disproportionately affects a person's decision to enroll in graduate school.

The cost parameter, $Cost_t$ is the average tuition and required fees for graduate and undergraduate students, both adjusted in terms of 2021 dollars (NCES,2023). For the model, there are two cost parameters, $Gradcost_t$ and $UndergradCost_t$. Note that cost is the total average between both public and private institutions. This would allow us to analyze how the increase in cost would be effective for students wanting to pursue higher education.

The unemployment parameter, $Unemployment_t$ (FRED, 2025) is a measure of the yearly average unemployment rate. As presented in the EAB infographic (Section 2), unemployment has been cited to have a statistically significant correlation to undergraduate and graduate program enrollments.

Lastly, the real interest rate, RIR_t. It is used as a metric to examine purchasing power and inflation, which indicates the actual cost of financing higher education. Simultaneously, it also measures the opportunity cost of a student attending school rather than working, where the higher the interest rate, the higher the cost of investing time by being in school. We use real interest rates in the model to represent willingness to take on student loans, which could theoretically impact post-secondary education enrollment rates.

3.3 Limitations of the model

While the model given by Equation 1 includes many factors of macroeconomy, there are some parameters that are omitted due to a lack of sufficient data, which could potentially lead to an omitted variable bias. Firstly, the cost parameter does not consider the aid that students receive. For low-income students, receiving grants, loans, and scholarships could significantly motivate them to enroll or not enroll in the University. Secondly, while unemployment is a

sufficient measure of how the state of the job market could encourage people to enroll in high-level education, the intended variable was the job opening-to-unemployment ratio to accurately represent the opportunity costs of attending school. There were only ten observations for this metric; therefore, it was omitted due to the small sample size. Lastly, the population of young adults could play into enrollment. For undergraduate students, the typical age range is 18-24, with small enough outliers to be negligible. On the other hand, there is no equivalent range to be used for graduate students because the age variance of graduate students is higher than undergraduate students. This may be because while a proportion of students go straight into graduate programs after finishing their undergraduate education, some choose to work before going back to school. Therefore, while the proposed model (Equation 1) has satisfactory explanatory power, the issues that may arise from the omitted variables are not negligible.

4. Empirical Results

4.1 Descriptive Statistics

The following table includes the number of observations, mean, standard deviations, minimum, and maximum of all the variables used in the regression. All parameters have 38 observations besides the graduate and undergraduate costs, with 35 and 37 observations, respectively.

Table 2: Descriptive statistics of all variables

Variable	**Name**	**Observation (N)**	**Mean**	**Standard Deviation**	**Minimum**	**Maximum**
Graduate Enrollment (in thousands)	gradenroll	38	116.59	18.40081	83.41	154.36
Undergraduate Enrollment (in thousands)	ugenroll	38	14688	2346.259	10798	16767
Graduate Cost (2021 dollars)	gradcost	35	16301	3980.57	9184	21703
Undergraduate costs (2021 dollars)	undergradcost	37	7663	3904.361	2312	14688
GDP per Capita Rate of Change	capitarate	38	4.088	2.417102	-2.830	10.720
GDP per Capita Rate of Change (squared)	capitasq	38	22.404	22.184	1.166	114.918
Unemployment rate	unemployment	38	5.792	1.568849	3.600	9.600
Real Interest Rate	RIR	38	2.0406	1.41925	-0.1195	4.4529

4.2 Population Models

4.2.1 Graduate Enrollment Population Model

Equation 2: Unrestricted Model for Graduate Enrollment

$$Grad\ Enrollment_t = \beta_0 + \beta_1 GDP_t^2 + \beta_2 GDP_t + \beta_3 Cost + \beta_4 Unemployment_t + \beta_5 RIR + \epsilon_t$$

Table 3: Summary statistics of Graduate Enrollment

	Coefficient Estimate	Standard Error	t-value	p-value
(Intercept)	120.760068	29.681975	4.068	0.000332 ***
capitasq	0.323333	0.111544	2.899	0.007069 **
capitarate	-1.417064	1.117918	-1.268	0.215028
gradcost	0.001106	0.001163	0.951	0.349461
unemployment	-1.317772	1.049407	-1.256	0.219238
RIR	-7.169619	3.619851	-1.981	0.057188 .

. p<0.1,*p < 0.05, **p < 0.01, ***p <0.001

The statistics presented in Table 2 suggest that there is strong evidence that GDP per capita change is correlated with graduate program enrollment (at 1% significant level). This validates the claim that graduate enrollment is indeed nonlinear, where extreme booms and busts increase enrollment. Secondly, there is some moderate evidence (at a 10% significance level) to suggest a correlation between real interest rates and graduate enrollment. This suggests that when interest rates rise, borrowing becomes more expensive, which potentially reduces willingness to take on student loans. Lastly, the adjusted R^2 for the model is 0.802.

4.2.2 Undergraduate Enrollment Population Model

Equation 3: Unrestricted Model for Undergraduate Enrollment

$$Undergrad\ Enrollment_t = \beta_0 + \beta_1 GDP_t^2 + \beta_2 GDP_t + \beta_3 Cost + \beta_4 Unemployment_t + \beta_5 RIR + \epsilon_t$$

Table 4: Summary statistics of Undergraduate Enrollment

	Coefficient Estimate	Standard Error	t-value	p-value
(Intercept)	10781.0421	1718.3746	6.274	5.64e-07 ***
capitasq	-21.2030	10.6200	-1.997	0.054724 .
capitarate	128.2927	102.2776	1.254	0.219088
gradcost	0.4003	0.1005	3.984	0.000382 ***
unemployment	309.5436	96.7035	3.201	0.003157 **
RIR	-525.9250	267.1129	-1.969	0.057957 .

. p<0.1,*p < 0.05, **p < 0.01, ***p <0.001

In Table 4, the undergraduate enrollment is statistically correlated with the macroeconomic parameters to varying degrees. GDP per capita rate of change squared (capitasq) and real interest rates (RIR) are negatively correlated with undergraduate enrollment at a 10% significance level. Unemployment is positively correlated with undergraduate enrollment at 1% significance level. Finally, average undergraduate tuition and required fees are positively

correlated with enrollment, at a 0.1% significance level. The adjusted R^2 for the regression is 0.9233.

4.3 Analysis of Graduate and Undergraduate Enrollment Models

From the analyses of the two unrestricted models, both graduate and undergraduate enrollments are correlated with GDP per capita rate squared (capitasq) to varying degrees. While graduate enrollments and GDP per capita rate squared were positively correlated at 1% significance level, undergraduate enrollments were negatively correlated with the independent variable at a 10% significance level. This indicates more evidence for the correlation between graduate enrollment and sensitivity to economic changes, wherein there are higher enrollments in times of rapid economic growth and decline, and lower enrollments in more stable economic conditions, as demonstrated by the positive quadratic relationship in Figure 1. Simultaneously, evidence suggests, at a 10% significance level, that the same booms and busts have a negative quadratic relationship with undergraduate enrollment trends, as observed in Figure 2. This implies that during times of high economic rate of change, undergraduate enrollments are low, and in times of stable economic conditions, undergraduate enrollments are higher.

Besides the GDP per capita rate of change variable, other variables demonstrated notable relationships between enrollment and macroeconomic trends. Such that undergraduate enrollments are positively correlated with undergraduate cost and unemployment at 0.1% and 1%, respectively; meanwhile, neither of these variables was correlated with graduate enrollment. This suggests strong evidence that undergraduate enrollments are sensitive to the costs and unemployment. This sensitivity implies that in times of high unemployment, people tend to see attending undergraduate programs as an option to potentially increase their employability. Furthermore, the positive relationship between undergraduate cost and undergraduate enrollment

may be due to the influence of financial aid availability. As costs increase, institutions and governments often allocate resources to help cover the costs, which would inherently increase access to undergraduate attendance for students and thereby, undergraduate enrollment. Although financial aid data is unavailable for the model due to the lack of available data, the positive relationship between cost and undergraduate enrollment may be partially capturing the effects of financial aid availability.

Lastly, both graduate enrollment and undergraduate enrollment are negatively correlated with real interest rates, which is significant at 10%. There are two potential explanations for the negative relationship. Firstly, when interest rates are high, it increases the cost of borrowing for education, especially when the government is allocating less spending to tuition assistance (TIAA, 2025). Therefore, when the cost of borrowing is high, people are discouraged from enrolling. Secondly, when interest rates are high, there is a higher opportunity cost of attending school rather than working. Therefore, individuals may be more inclined to join the workforce and/or save money rather than invest in an education.

The analyses of the graduate enrollment and undergraduate enrollment models show that undergraduate enrollment is correlated with more macroeconomic factors than graduate enrollment. This implies that the demand for graduate education may be more inelastic than the demand for an undergraduate degree.

4.4 Examining Multicollinearity

The variables that describe the macroeconomic landscape, such as cost and real interest rates, will naturally be correlated with each other. However, as demonstrated in the following

auxiliary regressions, while there is multicollinearity, there is no perfect collinearity between any two independent variables.

4.4.1 Graduate Program Cost Auxiliary

Equation 4 Graduate Cost Auxiliary Regression:

$$GradCost_t = \delta_0 + \delta_1 GDP_t + \delta_2 Unemployment_t + \delta_3 RIR_t + \epsilon_t$$

Table 5: Summary Statistics of Graduate School Cost Auxiliary Regression

	Estimate	Standard Error	t-value	p-value
(Intercept)	24991.14	1160.92	21.527	< 2e-16 ***
GDP per Capita (percentage change)	-201.96	95.21	-2.121	0.04199 *
Unemployment	-430.32	145.82	-2.951	0.00599 **
RIR	-2922.64	153.67	-19.019	< 2e-16 ***

. p<0.1,*p < 0.05, **p < 0.01, ***p <0.001

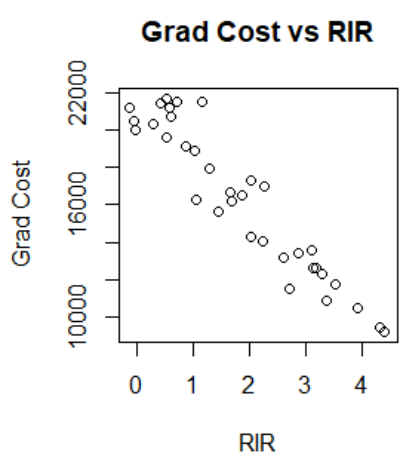


Figure 3 (left): Negative Linear Relationship between Graduate School Cost and Real Interest Rates

Figure 4 (right): Negative Linear Relationship between Graduate School Cost and Unemployment Rates

The graduate cost auxiliary regression demonstrates that there is strong evidence to suggest that there is a correlation between real interest rates and graduate school cost (Figure 3) and unemployment and graduate school cost (Figure 4). The low real interest rates indicate low borrowing costs, which may be indicative of an increase in the demand for graduate school, which increases graduate costs. Additionally, the countercyclical relationship between graduate cost and unemployment may also indicate that during recessions, more people want to enroll in graduate programs. This hypothesis is supported by the Education Advisory Board's infographic presented in the literature review (Section 2).

4.4.2 Undergraduate Program Cost Auxiliary

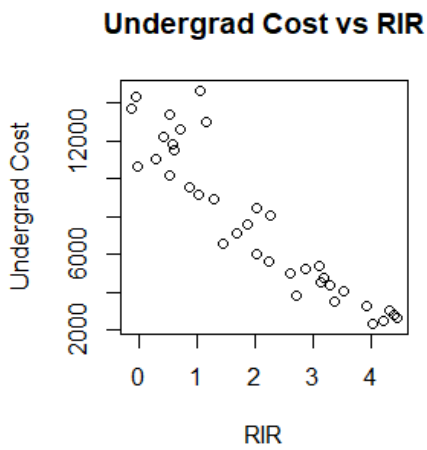
Equation 5: Undergraduate Cost Auxiliary Regression

$$UndergradCost_t = \delta_0 + \delta_1 GDP_t + \delta_2 Unemployment_t + \delta_3 RIR_t + \epsilon_t$$

Table 6: Summary Statistics of Undergraduate School Cost Auxiliary Regression

	Estimate	Standard Error	t-value	p-value
(Intercept)	15028.9	1207.7	12.445	5.2e-14 ***
GDP per Capita (percentage change)	135.3	100.4	1.347	0.1871
Unemployment	-421.4	154.9	-2.721	0.0103 *
RIR	-2655.4	148.1	-17.931	< 2e-16 ***

. p<0.1,*p < 0.05, **p < 0.01, ***p <0.001



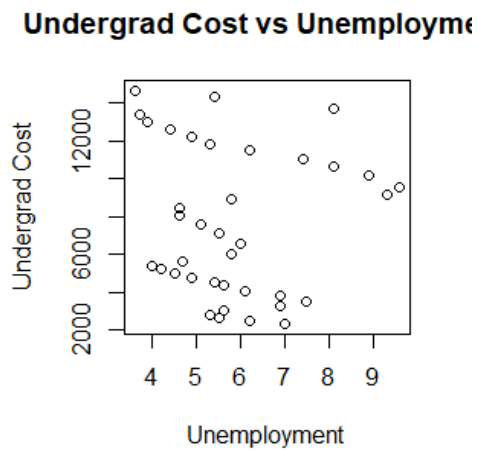


Figure 1: Negative Correlation between Undergraduate School Cost and Real Interest Rates

Figure 2: Negative Correlation between Undergraduate School Cost and Unemployment Rates

As demonstrated in Table 6, the undergraduate cost and real interest rate, and the undergraduate cost and unemployment rates are correlated at 0.1% and 10% significance level, respectively. The undergraduate cost, being correlated to the same variables as the graduate cost, reveals that the explanation for the correlation is similar.

4.5 Joint Significance

When examining the multivariate regression, the joint significance test (F-test) allows us to observe whether two variables are equally significant to enrollment. In this particular model, the variance inflation factor (VIF) is high for real interest rate and unemployment, and GDP per capita rate of change and unemployment, which makes their joint significance worth testing.

4.5.1 Joint Significance between Real Interest Rates and Unemployment

Null Hypothesis: $H_0: \beta_4 = \beta_5$ Alternate Hypothesis $H_a: \beta_4 \neq \beta_5$

Equation 6: Restricted Model for Joint Significance Testing between RIR and Unemployment

$$Enrollment_t = \beta_0 + \beta_1 GDP_t^2 + \beta_2 GDP_t + \beta_3 Cost_t + \beta_4 (Unemployment_t + RIR_t) + \epsilon_t$$

Table 7: Statistics for real interest rates and unemployment joint significance hypothesis testing

Model	**F-statistic**	**P-value**	**Conclusion at 5% significance level**
Graduate Enrollment	4.155	0.05041.	Fail to Reject the Null Hypothesis
Undergraduate Enrollment	11.928	0.001578***	Reject the Null Hypothesis

. p<0.1, *p < 0.05, **p < 0.01, ***p <0.001

The joint significance test was run for both graduate enrollment and undergraduate enrollment; the F-statistic and p-value are presented in Table 7. The results for the two joint significance tests differ for graduate enrollment versus undergraduate enrollment. In the graduate enrollment model, real interest rates and unemployment are equally significant at 5% significance level, whereas in the undergraduate enrollment model, real interest rates and unemployment are not equally significant at 5% significance level. It is important to note that these two variables are still significant to varying degrees, 10% in the graduate model and 0.1% in the undergraduate model. This indicates that they still influence undergraduate enrollment, but their combined explanatory effect is weaker than in the graduate model. Therefore, the contrast between joint significance reveals that real interest rates and unemployment jointly shape graduate enrollment decisions, whereas unemployment is the dominant explanatory variable in the undergraduate context.

4.5.2 Joint Significance between GDP per Capita and Unemployment

Null Hypothesis: $H_0: \beta_1 = \beta_4$ Alternate Hypothesis $H_a: \beta_1 \neq \beta_4$

Equation 7: Restricted Model for Joint Significance Testing between GDP per Capita squared and Unemployment

$$Enrollment_t = \beta_0 + \beta_1(GDP_t^2 + Unemployment_t) + \beta_2 Cost_t + \beta_4 RIR_t + \epsilon_t$$

Table 8: Statistics for GDP per Capita Rate squared and unemployment joint significance hypothesis testing

Model	**F -statistic**	**P-value**	**Conclusion at 5% significance level**
Graduate Enrollment	2.3792	0.1338	Fail to Reject the Null Hypothesis
Undergraduate Enrollment	10.686	0.002644**	Reject the Null Hypothesis

. p<0.1, *p < 0.05, **p < 0.01, ***p <0.001

After running the joint significance test, we found that unemployment and the squared GDP per capita rate of change variable do not have equal explanatory power in undergraduate enrollment (p=0.00264) as they do in graduate enrollment (p=0.1338). It is important to note that when running the joint significance test for the linear GDP per capita variable, we fail to reject the null hypothesis for both at 5% significance level with 0.9335and 0.08347 as the p-value for graduate enrollment and undergraduate enrollment, respectively. The discrepancy can be explained by the fact that undergraduate enrollments are more sensitive to both nonlinear swings in the economy and unemployment rates, as established in Section 4.2.2.

5. Conclusion

The multivariable regression analyses of graduate enrollment and undergraduate enrollment yielded several meaningful insights. First, during model specification, we identified a positive quadratic relationship between GDP per capita and both undergraduate and graduate

programs (Figure 1 and Figure 2). The quadratic nature implies high enrollments when the GDP per capita rate of change is in rapid growth and decline. Furthermore, we discussed the variables left out in the model due to the lack of data, which could lead to omitted variable bias in the model.

Examining the coefficient estimates of graduate enrollments and undergraduate enrollments in unrestricted models (Table 1 and Table 2). The statistical analyses of the graduate and undergraduate enrollment unrestricted model revealed that graduate enrollment correlated with economic swings at a higher statistical significance level than undergraduate enrollment. However, undergraduate enrollments are correlated with more macroeconomic parameters than graduate enrollments. Therefore, we were able to conclude that graduate enrollments are less responsive to changes in the macroeconomic landscape than undergraduate enrollments, consistent with the notion that demand for graduate education may be more elastic. Meanwhile, undergraduate enrollments appear more sensitive to labor market conditions, borrowing costs, and economic variability.

Subsequently, the auxiliary regressions showed that both graduate and undergraduate costs were extremely correlated with real interest rates (4.4.1, 4.4.2). Finally, the joint significance test was conducted between real interest rates and unemployment, as well as real interest rates and GDP per capita, to assess if each variable was equally important as the other. As demonstrated in Tables 7 and 8, at the 5% significance level, the null hypothesis could not be rejected for graduate enrollment, whereas it was rejected for undergraduate enrollment. Thus, for undergraduate enrollments, real interest rates, unemployment, and GDP per capita have statistically different explanatory effects; while, in contrast, graduate enrollment appears more resistant to such macroeconomic changes.

These findings have several implications for higher education policy and planning. As observed throughout the data analyses of graduate enrollment, the demand for a graduate degree is less sensitive to macroeconomic changes. Institutions can work to stabilize enrollment by reducing tuition volatility. In the meantime, as undergraduate enrollments are sensitive to macroeconomic shifts, federal and local policymakers should consider allocating more funding to expand financial aid to undergraduate students, whether through grants. scholarships, subsidized loans, or tuition assistance. This would ensure that those who intend to pursue higher education in a labor market where a degree is the new natural resource gain access and the help they need to successfully achieve their goals. As previously summarized from TIAA, pursuing a higher education not only benefits the individual but also the community, and consequently, the broader economy.

References

Bureau of Labor Statistics. (2024). *61.4 percent of recent high school graduates enrolled in college in October 2023.* https://www.bls.gov/opub/ted/2024/61-4-percent-of-recent-high-school-graduates-enrolled-in-college-in-october-2023.htm

Deloitte. (2023). *Trend No.1: College enrollment reaches its peak.* https://www.deloitte.com/us/en/insights/industry/articles-on-higher-education/enrollment-rates-in-higher-education.html

Education Advisory Board. (2016). *How to understand higher education enrollments in economic recessions.*https://eab.com/resources/infographic/understand-higher-education-enrollments-economic-recessions/

FRED. (2025). *Unemployment Rate (UNRATE)* https://fred.stlouisfed.org/series/UNRATE

FRED. (2025). *10-Year Real Interest Rate (REAINTRATREARAT10Y)***.** https://fred.stlouisfed.org/series/REAINTRATREARAT10Y

Georgetown University Center on Education and the Workforce. (2021). *Projections of Jobs, Education, and Training Requirements through 2031.* https://cew.georgetown.edu/cew-reports/projections2031/#:~:text=The%20blue%2Dcollar%20and%20skilled%2Dtrades%20economy%20consists%20of%20the,high%20school%20diploma%20or%20less.

Hanson, Melanie. (2025) *Education Attainment Statistics*. Education Data Initiative. https://educationdata.org/education-attainment-statistics

NCES. (2022*). Average undergraduate tuition, fees, room, and board rates charged for full-time students in degree-granting postsecondary institutions, by level and control of institution:*

Selected years, 1963-64 through 2020-21

https://nces.ed.gov/programs/digest/d21/tables/dt21_330.10.asp

NCES. (2022*). Average and percentiles of graduate tuition and required fees in degree-granting postsecondary institutions, by control of institution: Academic year 1989-90 through 2021-22*. https://nces.ed.gov/programs/digest/d22/tables/dt22_330.50.asp

NCES. (2022). Table 303.70 Total undergraduate fall enrollment in degree-granting postsecondary institutions, by attendance status, sex of student, and control and level of institution: Selected years, 1970 through 2030 https://nces.ed.gov/programs/digest/d21/tables/dt21_303.70.asp

NIH (2025). National Statistics on Graduate Students. https://report.nih.gov/nihdatabook/category/20

TIAA Institute (2025). *Macro-trends impacting the evolution of higher education: How colleges and universities can adapt to create value in the 21st century.* https://www.tiaa.org/public/institute/publication/2025/macro-trends-impacting-evolution-of-higher-education

World Bank. (2025) *GDP per capita growth (annual %)* https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG